Qianxun (Cecilia) Xu

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EDUCATION

University of California, Los Angeles 2026 - 2028

- M.S. in Computer Science

Duke Kunshan University (DKU) & Duke University Dual Degree 2022 - 2026

- B.S. in Applied Math and Computer Science; Computer Science Track.
- GPA: 3.93/4.0; Major GPA: 3.97/4.0
- Awards: IEEE ICPADS Best Presentation in Session Award, NeurIPS Scholar Award ($2,000), DKU Merit-based College Entrance Scholarship ($48,000).

PUBLICATIONS

Xu, Q., Song, C., Cai, Y., Zhang, C. SwitchCraft: Training-Free Multi-Event Video Generation with Attention Controls. *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (**CVPR**), 2026.*

Xu, Q., Boukerche, A., Sun, P. FENSe: Feedback-Enabled Neighbor Selection for Spatial Aware Collaborative Perception. *IEEE International Conference on Parallel and Distributed Systems (**ICPADS**), oral, 2025.*

Xu, Q., Li, Z. Partial Physics Informed Diffusion Model for Ocean Chlorophyll Concentration Reconstruction. *Advances in Neural Information Processing Systems (**NeurIPS**), main track poster, 2025.*

RESEARCH EXPERIENCE

Training-Free Multi-Event Video Generation May - Nov 2025

Supervisor: prof. Chi Zhang, Westlake University Hangzhou, China

- **Problem:** Addressed semantic confusion, temporal misalignment, and high fine-tuning costs in existing text-to-video (T2V) diffusion models when generating multi-event videos.
- **Method:**
  * **Event-Aligned Query Guidance:** Extracted text anchors to construct event-specific Key subspaces. Projected frame Queries with bidirectional guidance (enhancing targets, suppressing competitors) for strict temporal alignment.
  * **Adaptive Strength Solver:** Designed a convex optimization solver to dynamically allocate guidance intensity based on projection gaps, balancing controllability and generation quality.
- **Results:** Achieved SOTA text-to-video alignment scores while preserving temporal smoothness and visual quality on the Wan 2.1 14B baseline, providing an effective and efficient solution for multi-event video generation.

Feedback-Enabled Neighbor Selection for Autonomous Vehicles, *DKU Undergraduate Thesis* Jan - Aug 2025

Supervisor: prof. Peng Sun, DKU Kunshan, China

- **Problem:** Addressed redundant feature transmission and high communication latency from highly overlapping fields of view in V2X collaborative perception.
- **Method:** Grouped overlapping vehicles via traffic-aware dynamic clustering. Leveraged low-bandwidth feedback signals to evaluate cluster representatives, selectively requesting high-dimensional features to eliminate redundancy.
- **Results:** Achieved superior accuracy-efficiency trade-off. Improved detection accuracy by ~4% under equal bandwidth while significantly reducing ego-side computation time.

Partial Physics Informed Diffusion Model May 2024 - May 2025

Supervisor: prof. Zuchuan Li, DKU Hangzhou, China

- **Problem:** Addressed sparse data imputation in environmental monitoring, overcoming poor generalization in data-driven models and rigid mechanism modeling in traditional physics-based methods.
- **Method:**
  * **Selective Physics Encoding:** Designed selective encoding mechanism using observable advection dynamics as core constraints. Prevented unobservable biological terms from misleading training signals.
  * **Subspace Alignment Loss:** Formulated physics subspace alignment loss. Aligned predictions with ground truth exclusively within the subspace of known physical dynamics, bypassing forced full-equation residuals for unknown parameters.
- **Results:** Achieved robust long-range spatiotemporal inference. Reduced reconstruction RMSE by 34% against pure data-driven models, and by 49% against baselines forcing complete physical equations.

PROJECT EXPERIENCE

TransGCN For Muscle Activation Prediction from Joint Angles During Motion Mar - May 2025

- Adapted a spatio-temporal Transformer with a kinematic graph convolutional network to predict frame wise muscle activations from joint angle sequences. Integrated demographic embeddings (gender/weight/height) for personalized estimates.

CAMPUS EXPERIENCE

Computer Science 101 Course Teaching Assistant Oct - Dec 2022

- Led weekly lab sessions and office hours; graded quizzes and assignments; analyzed error trends and common confusions.
- Utilized data-driven insights to recommend curriculum adjustments, increasing student engagement and academic achievement.

ADDITIONAL INFORMATION

Languages: English (GRE:324/340+4.5/6; TOEFL:116/120), German (intermediate), Mandarin (native).

Skills: Python, Java, C, PyTorch, Git, Linux, Android Studio.